

Education

- PhD** in Computer Science (Computer Graphics), **New York University** September 2020 - August 2025
- Thesis: *Computational Design through Differentiable Elastodynamic Simulation, Parametrized by Geometric Techniques, for Applications in Soft Robotics*
 - Advisors: Daniele Panozzo and Denis Zorin, Geometry Computing Lab, GPA 3.9
- BA** in Physics and Computer Science, **New York University** September 2014 - May 2018
- Honors: Phi Beta Kappa, Society of Physics Students, GPA 3.9

Experience

- New York University, New York, NY** – Graduate Research Assistant September 2020 - August 2025
- Created one of the first comprehensive benchmarks for assessing the unbiased performance of Scientific Machine Learning methods on dynamical systems.
 - Formulated and developed a high performance, differentiable (shape, material, boundary conditions) Finite Element simulator for use in inverse design nonlinear elastic problems with contact, surpassing existing methodologies in scope and functionality (comparison in paper).
 - Created optimization methodologies for computational design of soft robots and deformable, capacitive stretch sensors using differentiable FEM elastic and electrostatic simulation. Validated these methodologies through physical fabrication and experimental design, resulting in strong qualitative and quantitative improvements.
 - Researched, prototyped and optimized a novel PCB-based tactile sensing methodology for use in robotics, tracking shear, torque and pressure.
 - Managed and mentored Masters students involved in research projects.
 - Course Assistant: Computer Graphics, Geometry Processing
 - Reviewer: NeurIPS (2022, 2023, 2024), SIGGRAPH (2025)
- NTop, New York, NY** – Software Engineer Intern June - August 2025
- Integrated PolyFEM simulation and optimization software into NTop's design software for nonlinear and time-dependent analysis, targeting specific functionality needs from customers for advanced engineering design projects.
- Disney, Zurich, CH** – Research Intern June - August 2022
- Engineered a patented framework to enable optimization of soft elastomeric skins draped over humanoid robots to match artistic intent.
 - Developed a finite element method based optimization targeting artistic designs in pneumatic soft robotics and validated approach via fabrication, leading to SIGGRAPH publication (collaboration until May 2024).
- Google, New York, NY** – Software Engineer September 2019 - September 2020
- Engineered pipelines of vision models for on-device long tailed classification tasks and worked on long tailed deep learning models for academic use (Fine Grained Visual Classification, Google Research)
- Google, New York, NY** – Engineering Resident September 2018 - September 2019
- Improved transit prediction quality through feature engineering on large geospatial data (Google Maps)
 - Conducted study on vision performance of deep learning methods on bird species classification (Google Research)
- VIDA Lab at NYU Tandon, Brooklyn, NY** – Research Assistant June - August 2018
- Created a human-in-the-loop approach to generate baseball position tracking using historical data, resulting in CHI publication and patent
- Entrupy, New York, NY** – Computer Vision Intern June - August 2017
- Developed classical and data-driven vision algorithms on surface textures (leather, cloth), analyzing failure modes

Skills

- **C++** (7 years, Eigen, libigl), **Python** (10 years, PyTorch, SciPy suite, OpenCV)
- Contributor to PolyFEM, a differentiable C++ finite element simulator
- Experience in Path Tracing and Rasterization Algorithms, Geometry Processing, Computer Vision and Machine Learning Techniques, Numerical Methods for Differential Equations, Microelectronics, Additive Manufacturing

Publications

Computational Modeling and Design of Capacitive Stretch Sensors

Arvi Gjoka, Yongkang Sun, Roi Poranne, Daniele Panozzo
ACM Transactions on Graphics (SIGGRAPH Asia), 2025

Soft Pneumatic Actuator Design using Differentiable Simulation

Arvi Gjoka, Espen Knoop, Moritz Bächer, Denis Zorin, Daniele Panozzo
SIGGRAPH 2024

Differentiable solver for time-dependent deformation problems with contact

Zizhou Huang*, Davi Tozoni*, Arvi Gjoka, Zachary Ferguson, Teseo Schneider, Daniele Panozzo, Denis Zorin
ACM Transactions on Graphics (SIGGRAPH), 2024

An Extensible Benchmark Suite for Learning to Simulate Physical Systems

Karl Otness, Arvi Gjoka, Joan Bruna, Daniele Panozzo, Benjamin Peherstorfer, Teseo Schneider, Denis Zorin
NeurIPS, 2021

The iWildCam 2020 Competition Dataset

Sara Beery, Elijah Cole, Arvi Gjoka
CVPR, 2020, Fine-Grained Visual Categorization Workshop

HistoryTracker: Minimizing Human Interactions in Baseball Game Annotation

Jorge Piazzentin Ono, Arvi Gjoka, Justin Salomon, Carlos A. Dietrich, Cláudio T. Silva
CHI, 2019